



A COMMUNITY OF CHOICE

Planning and Development Department

2101 O'Neil Avenue, Suite 202, Cheyenne WY 82001
(Phone) 307-637-6282 (Fax) 307-637-6366

MEMO

To: Honorable Mayor Patrick Collins and Members of the City Council
From: Charles W. Bloom, AICP
Subject: UDC-23-00339 | Ironhorse 3rd Filing Outside User Agreement
Date: May 22, 2023

Background:

The item for consideration is an *Outside User Agreement*. This is approved by Resolution.

The proposed Governing Body review schedule is as follows:

- Introduction May 22, 2023
- Public Services Committee June 6, 2023
- Consideration June 12, 2023

Please see that attached staff report for additional information.

Staff Recommendation:

Staff recommends the Governing Body not approve the resolution authorizing the outside user agreement.

Attachments:

1. Resolution
2. Staff Report including:
 - a. BOPU flow calculation worksheet
 - b. County Staff Report
 - c. University of Wyoming Center for Business and Economic Analysis study "Evaluating the Economic Impact of Providing Municipal Water to Laramie County Housing Developments"

RESOLUTION NO. _____

ENTITLED: "A RESOLUTION APPROVING WATER SERVICE, BY OUTSIDE USER AGREEMENT, FOR PROPERTIES TO BE KNOWN AS IRONHORSE 3RD FILING, LARAMIE COUNTY, WYOMING GENERALLY LOCATED SOUTHEAST OF WEST RIDING CLUB ROAD AND HYND'S BOULEVARD."

WHEREAS, Ironhorse Capital, Limited Partnership, is the owner of four (4) properties in Laramie County, Wyoming, to be known as Ironhorse 3rd Filing, Laramie County, Wyoming more specifically described as a ±97.93-acre portion of the NE1/4 of Section 12, Township 14 North, Range 67 West of the 6th P.M., Laramie County, Wyoming; more particularly described as:

All of Tract 11, Sunset Tracts, 3rd Filing; along with the adjacent portion of West Ole Maverick Road Right-of-Way (40');

Tract 12, Sunset Tracts, 3rd Filing LESS the East 15 feet of Tract 12 deeded to the State Highway Commission of Wyoming by Warranty Deed dated October 18, 1968, recorded August 4, 1969 in Book 895, Page 450; along with that adjacent portion of West Ole Maverick Road Right-of-Way (40');

All of Tracts 1 & 2, Ironhorse, 1st Filing;

All of Tracts 3 & 4, Ironhorse, 2nd Filing;

All of that unplatted parcel situated in the S1/2S1/2NE1/4 of said Section 12; a/k/a "Parcel I" in Deed Book 2659, Page 1542, and described in said Deed as:

Beginning at a point of the East WYDOT right of way which lies S00°13'30"E a distance of 1936.78 Feet from a point of intersection of the West WYDOT right of way and South right of way of West Riding Club Road; Thence N89°52'38"E a distance of 2482.13 Feet to the point on the West right of way of Yellowstone Road; Thence along said West right of way of Yellowstone Road S00°00'36"W a distance of 658.88 Feet to a point on the North boundary of Sunset Tracts First Filing; Thence along said North boundary of Sunset Tracts First Filing S89°52'01"W a distance of 2479.43 feet to a point on the East WYDOT right of way; Thence along said East WYDOT right of way N00°13'30"W a distance of 659.33 Feet more or less to the Point Of Beginning; ALONG WITH that adjacent portion of Buckbrush Road Right-of-Way (40'); and,

All of that remaining unplatted portion of that parcel situated in the S1/2NW1/4NE1/4 and the N1/2SW1/4NE1/4 of said Section 12 AKA "Parcel II" in Deed Book 2659, Page 1542, and described in said Deed as:

Beginning at a point on the east WYDOT right of way which lies S.00°13'30"E. a distance of 619.35 feet from a point of intersection of the west WYDOT right of way and south right of way of West Riding Club Road; thence N.89°50'51"E. a distance of 1234.40 feet to a point on the west right of way of Buck Brush Road; thence along said west right of way of Buck Brush Road S.00°24'29"W. a distance of 1318.13 feet to a point; thence S.89°52'38"W. a distance of 1219.82 feet to a point on said east WYDOT right of way; thence along said east WYDOT right of way N.00°13'30"W. a distance of 1317.43 feet more or less to the point of beginning; ALONG WITH that adjacent portion of Buckbrush Road Right-of-Way (40');

WHEREAS, Ironhorse Capital, Limited Partnership, ("the Applicant") proposes to develop a clustered residential subdivision of approximately thirty-seven (37) lots which includes thirty-three (33) residential, three (3) open space, and one (1) vacant lot;

WHEREAS, Applicant submitted an application to the City of Cheyenne and the Board of Public Utilities ("BOPU") for water service outside of the corporate limits of the City of Cheyenne for the above-referenced properties;

WHEREAS, pursuant to Resolution No. 3577, BOPU was instructed to present to the Governing Body for its review any application for the provision of water and/or sanitary sewer services to any individual or entity outside the corporate limits of the City of Cheyenne;

WHEREAS, BOPU has reviewed the application submitted by the Applicant, and BOPU and the Applicant will approve and execute applicable Outside User Agreement(s) upon approval by the Governing Body pursuant to Resolution No. 3577; and,

WHEREAS, the Governing Body has reviewed the applications for water and/or sanitary sewer services submitted by the Applicant and has determined that approval is in the best interest of the City of Cheyenne.

NOW, THEREFORE BE IT RESOLVED BY THE GOVERNING BODY OF THE CITY OF CHEYENNE, WYOMING, THAT the application submitted by the Applicant to BOPU for water services outside of the corporate limits of the City of Cheyenne for the property described herein, is hereby approved,

BE IT FURTHER RESOLVED, that BOPU and the Applicant are authorized to approve and execute Outside User Agreement(s).

PRESENTED, READ AND ADOPTED this _____ day of _____,
2023.

Patrick Collins, Mayor

(SEAL)

ATTEST:

Kristina F. Jones, City Clerk



A COMMUNITY OF CHOICE

Planning and Development Department

2101 O'Neil Avenue, Suite 202, Cheyenne WY 82001

(Phone) 307-637-6282 (Fax) 307-637-6366

CITY OF CHEYENNE GOVERNING BODY

MAY 22, 2023

STAFF REPORT

FILE: UDC-23-00339: Ironhorse 3rd Filing Outside User Agreement

REQUEST: An outside user agreement for the provision of water for a clustered residential subdivision of approximately thirty-seven (37) lots which includes thirty-three (33) residential, three (3) open space, and one (1) vacant lot

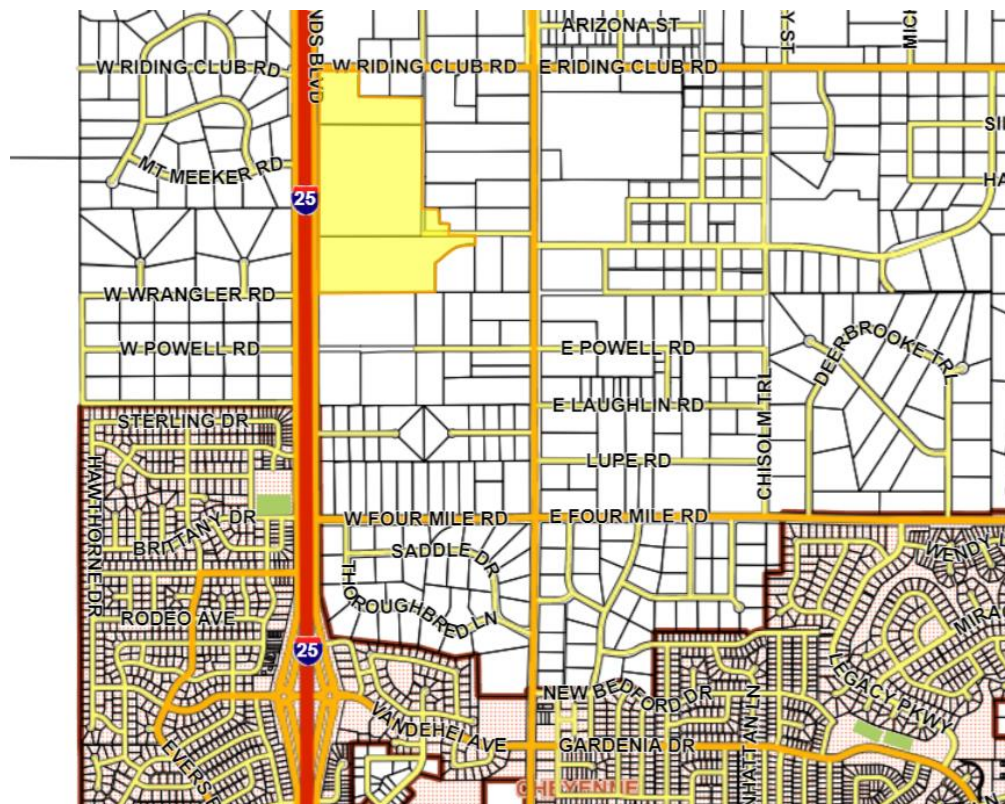
LOCATION: Southeast of the intersection of Happy Jack Road and Bruns Way

APPLICANT: Ironhorse Capital, Limited Partnership

AGENT: Casey Palma, Palma Land Planning

OWNERS: Ironhorse Capital, Limited Partnership

PREPARED BY: Charles W. Bloom, AICP, Planning and Development Director



APPLICABLE CITY CODE SECTION(S) AND PLANS:

- PlanCheyenne
- Resolution 3577

BACKGROUND:

The applicant is requesting connection to the BOPU water system for property outside City limits for a subdivision to be known as Ironhorse 3rd Filing, Laramie County, WY. The proposed subdivision is comprised of approximately 97.93-acres and proposes a clustered residential subdivision of approximately thirty-seven (37) lots which includes thirty-three (33) residential lots, three (3) open space lots, and one (1) vacant lot. The subdivision is approximately 1,300 feet north of City limits.

The County Planning Commission Staff Report is attached. The County Planning Commission considered this item on May 11, 2023, and recommended approval subject to conditions, including a condition that this outside user agreement was approved. The next step is for the applicant to submit a final plat for review by County Staff, the County Planning Commission, and the Board of County Commissioners.

For details regarding the development, please see the County Planning Commission Staff report which is attached.

ANALYSIS OF APPLICATION CONFORMANCE WITH REVIEW CRITERIA:

The request for an outside user agreement has no formal review criteria; however, the following elements should be considered when reviewing outside user agreements:

- BOPU capacity
- Needed infrastructure improvements
- Population served
- Considerations on the effect on future annexation(s)
- Fiscal cost/benefit to the City and the BOPU
- Conformance with the Comprehensive Plan

BOPU has identified that capacity exists to provide service to the subdivision. The applicant proposes constructing a private system that will be maintained by a private district. Typical water usage for a single-family home is approximately 6,000 per month and commercial development heavily depends on the end user. Attached is the Board of Public Utilities' (BOPU) flow calculation worksheet for this subdivision.

County development regulations require lots to have a minimum lot area of 5.25 acres. With this standard, the area within the plat boundary could support between 10 and 12 residential lots, depending on right-of-way dedication. Provision of an outside user agreement allows for the developer to develop an additional 21-23 single family homes and a potential commercial area. At present, the City has sufficient system capacity to provide that additional capacity, however granting of this agreement would decrease the ability of the City to allocate that water to properties within the City limits, whether needed or not.

The private system will be built to a rural system standard and will be maintained by the special district. The special district will be responsible for all maintenance and repairs. The proposed service line will be installed across lot frontages that are not in the district. In the event any other property owners seek to

connect to the private system, the district would have to be expanded and the system would have to be evaluated by the BOPU and Wyoming Department of Environmental Quality (WDEQ) standards. If annexed, the system would require upgrading to City standards.

In April of 2023, the Center for Business and Economic Analysis (CBEA) at the University of Wyoming evaluated the costs to the City for providing municipal water to County developments. The study was entitled "Evaluating the Economic Impact of Providing Municipal Water to Laramie County Housing Developments" (Study) and concluded that "for nearly all the parameter values, it does not make sense to provide municipal water to Laramie County new housing developments" (Study, p. 5).

4% sales tax revenue, 1% sales tax revenue, gas tax revenue, severance taxes, and State direct distribution are all dependent on a City's percentage of the overall population of the County. The Study concluded that for each additional person added to the incorporated County, the City lost \$626 per new resident. With the average household size being 2.29 persons per household, this translates into \$1,433.54 per household. This agreement would enable at least 21-23 additional units which translates into \$30,104.34 and \$ 32,971.42, respectively, annually for just the residential component of the subdivision. The Study's parameters looked at the cost to develop a ten-lot subdivision on a five-acre lot which would yield lots averaging approximately 21,780 sq. ft., which is just over twice as dense as the residential lot sizes proposed within Ironhorse 3rd Filing.

Property taxes would increase; however, the increased property tax would benefit the County and local fire district. BOPU would charge a higher rate for outside users (150% percent of City rate) and BOPU would not be required to maintain any of the infrastructure related to the special district – so this agreement would likely be a net benefit to BOPU.

Annexation of the property is not necessarily straightforward but could present some contiguity problems. If annexation did occur, City police, fire, and sanitation services would have to be extended well outside their current service boundary to service this subdivision likely resulting in a net fiscal loss. Annexation could be accomplished by a "flagpole annexation" or "annexation off a city island." If annexation is proposed, City staff will review the proposal to determine any necessary impact fees for public safety or other community services. The City would not entertain maintaining any park facilities or common areas within the subdivision.

Enabling establishment of lots smaller than five-acres does present planning challenges for future City growth. It is inherent that cities grow and at some time Cheyenne will begin to densify to the north. Lots smaller than five acres are generally challenging to redevelop. Lots ranging between one and five acres require significant infrastructure to support. A typical single-family lot street frontage is 60' and the proposed rural lots have frontages four to five times the typical width of a suburban lot. Single family home lots in the City struggle to come close to supporting ongoing maintenance costs for streets, utilities, and City support services. If this property is ever annexed to the City, the subdivision will not be able to financially support the maintenance and any necessary upgrades to infrastructure within and adjacent to the subdivision. When the incorporated City reaches this area, development of City-standard infrastructure extensions to "leapfrog" past this development would place a fiscal burden on a developer, or perhaps, the City.

CONCLUSION:

Staff is not satisfied that the application approval is in the best overall interests of the City due.

STAFF RECOMMENDATION:

Staff recommend denial of the proposed outside user agreement by not passing the associated resolution.

ATTACHMENTS:

1. BOPU flow calculation worksheet
2. County Staff Report
3. University of Wyoming Center for Business and Economic Analysis study "Evaluating the Economic Impact of Providing Municipal Water to Laramie County Housing Developments"

Applicants must use this worksheet to calculate flows and submit to Cheyenne BOPU using the email link: bdorr@cheyennebopu.org

Required Fire Flow:	TBD	gpm
----------------------------	-----	-----

[Link to IPDC](#)

¹ 2.5 people per residence (U.S. Census Bureau)

Please use the table below to calculate water and sewer flows for your project.

Sewer	Avg. Flow (gpm)	Max Day Flow (gpm)	Peak Flow (gpm)
	4.9	9.9	27.7
	0.0	0.0	0.0
	0.0	0.0	0.0
	0.0	0.0	0.0
	0.0	0.0	0.0
	0.0	0.0	0.0
Sub-Totals:	4.9	9.9	27.7

Please use the table below to calculate water and sewer flows for any future phases of your project.

Future Sewer	Avg. Flow (gpm)	Max Day Flow (gpm)	Peak Flow (gpm)
	0.0	0.0	0.0
	0.0	0.0	0.0
	0.0	0.0	0.0
	0.0	0.0	0.0
	0.0	0.0	0.0
Sub-Totals:	0.0	0.0	0.0

4.9	9.9	27.7
-----	-----	------

Please use the table below to identify tie-in locations and associated flows to existing system. Provide BOPU water/sewer main or manhole ID when applicable.

Email: casey@plandesignwy.com



LARAMIE COUNTY PLANNING & DEVELOPMENT DEPARTMENT

Planning • Building

MEMORANDUM

TO: Laramie County Planning Commission

FROM: Justin Arnold, Planning Program Manager

DATE: May 11, 2023

TITLE: PUBLIC HEARING regarding a review and recommendation of the Zone Change and Preliminary Development Plan for Ironhorse, 3rd Filing, located in a portion of the NE1/4 of Section 12, T.14N., R.67W., of the 6th PM, Laramie County, WY.

EXECUTIVE SUMMARY

Palma Land Planning, LLC, on behalf of Ironhorse Capital, has submitted a Preliminary Development Plan application to include a zone change to Low Density Residential (LR) for Ironhorse, 3rd Filing, located east of Hynds Blvd, west of Yellowstone Rd. and north of W. Wrangler Rd. The application has been submitted in order to prepare for clustered residential subdivision of approximately 37 lots, 33 residential, 3 open space and 1 vacant lot.

BACKGROUND

There are (4) properties that make up Ironhorse, 3rd Filing all owned by Ironhorse Capital Limited Partnership. The contiguous area is vacant with no structures. The surrounding area consists of other residential lots ranging in size from 2.78 to 9.78 acres.

On October 27th 2022 a Preliminary Development Plan, not including the current zone change proposal, for Ironhorse 3rd filing was brought forward to the Laramie County Planning Commission. The PDP was tabled (5-0) so the applicant could ascertain more detailed information for an outside water use agreement with the Cheyenne Board of Public Utilities. In order for the configuration of the current PDP to be considered, the applicant would need to have an outside water use agreement or a public water and sewer system in place to develop lot sizes that are less than 5.25 acres, per the AMEC memo and Development Design Standards. At this time a final decision regarding the outside user agreement has not been made, the applicant wishes to proceed with a public supply via a community well permitted by the State and regulated by the Environmental Protection Agency (EPA).

Pertinent Laramie County Land Use Regulations

Section 2-1-100 governing the requirements for submittal of a Preliminary Development Plan.

Section 1-2-103(b) governing required findings of the Planning Commission to recommend approval of a change in zone district.

Section 1-2-104(b)(D) governing notice to area property owners for a Preliminary Development Plan.

Section 4-2-103 governing the Low Density Residential (LR) Zone District

DISCUSSION

The Laramie County Comprehensive Plan identifies the area as Urban Rural Interface (URI). Within this area, higher density residential, intensive commercial, employment centers and industrial uses are indicated. Generally, these uses have access to public water and sewer services. Adjacent jurisdictions should be involved in decision making in the URI, this is important due to the potential water agreement with the City of Cheyenne. While some of these areas may include existing agricultural uses, it is anticipated they are more service-oriented and may change and develop over time.

PlanCheyenne contemplates Rural Residential uses for this property. Uses include single and multi-family residences, generally served by collector or arterial streets. Supporting and complementary uses, including open space recreation, schools, places of worship and other public or civic uses are encouraged. Residential densities vary depending on use and zoning.

According to the Order of the State Engineer for the Laramie County Control Area dated April 1, 2015, this property currently lies within the Conservation Area, in which the well spacing requirements for new permits are indicated at one (1) well for domestic use per each lawfully subdivided lot or one (1) per each ten (10) acres of any other legal parcel of land. A letter received from the State Engineer's Office, dated April 11, 2023 which is also attached herein, allows for wells to be permitted but strongly suggests that a monitor well be installed down gradient to allow for the aquifer impact of this subdivision's water usage to be studied.

A subdivision of more than 5 lots requires submittal of a Chapter 23 study to the Wyoming Department of Environmental Quality (DEQ). A letter received by the Laramie County Board of Commissioners on April 20, 2023 indicates provides an adverse recommendation for the referenced project meaning that the proposed subdivision does not meet the safety and adequacy requirements of DEQ Chapter 23 Rules as submitted, but that conditions listed in attachment 5 may be placed on the subject septic installations to mitigate impacts.

The subdivision will be accessed primarily from Hynds Blvd., an 80-foot right-of-way from Hynds Blvd to Yellowstone Road, being a two lane rural highway. Internal roadways will also provide for an 80-foot right-of-way according to the county engineer comments. A traffic study has been submitted with the application.

The Environmental and Services Impact Report submitted by the applicant concludes that the proposed development will not contribute to any measurable level of negative impacts on the area, and that several benefits will be realized for the stakeholders, agencies, rural county landowners and the public at-large.

Subsequent to October 27, 2022 Planning Commission meeting, Planning and Development has received several inquiries and letters in objection to the proposal, letters to which are contained within this package.

RECOMMENDATION and FINDINGS

Based on evidence provided, staff recommends the Planning Commission find that:

- a. This application meets the criteria for a preliminary development plan pursuant to section 2-1-100 of the Laramie County Land Use Regulations.

and that the Planning Commission may approve the Ironhorse, 3rd Filing, Preliminary Development Plan and Zone Change with the following conditions:

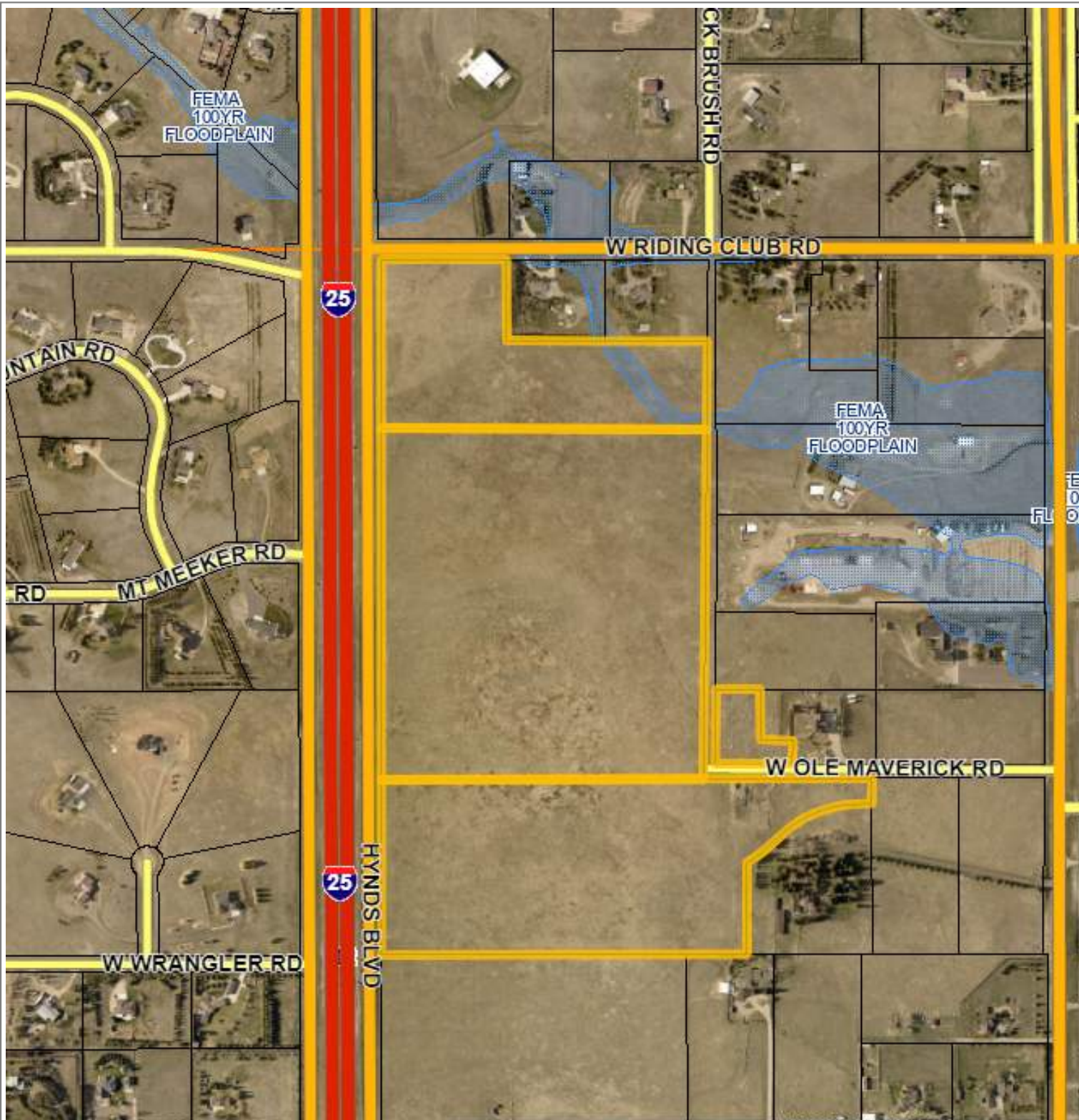
1. The applicant shall receive an approved outside user agreement from the City of Cheyenne prior to applying for a Zone Change with Subdivision Permit and Plat.
2. The applicant shall place notes stipulated by the Wyoming Department of Environmental Quality on the plat.

PROPOSED MOTION

I move to approve the Ironhorse, 3rd Filing, Preliminary Development Plan to include a Zone Change to Low Density Residential (LR) and adopt the findings of fact a of the staff report with two (2) conditions.

ATTACHMENTS

- Attachment 1: Location and Jurisdiction Map**
- Attachment 2: Preliminary Development Plan / Land Analysis Map**
- Attachment 3: State Engineer Letter**
- Attachment 4: Department of Environmental Quality Recommendation**
- Attachment 5: Agency Comments**
- Attachment 6: Public Comment Letters**



Aerial and Location Map

Preliminary Development Plan
Ironhorse, 3rd Filing
PZ-22-00294

Comprehensive Plan- Urban Rural Interface (URI)

AMEC- Zone 2

Zoning- AR (Agricultural Residential)

Fire District- Laramie County Fire Authority

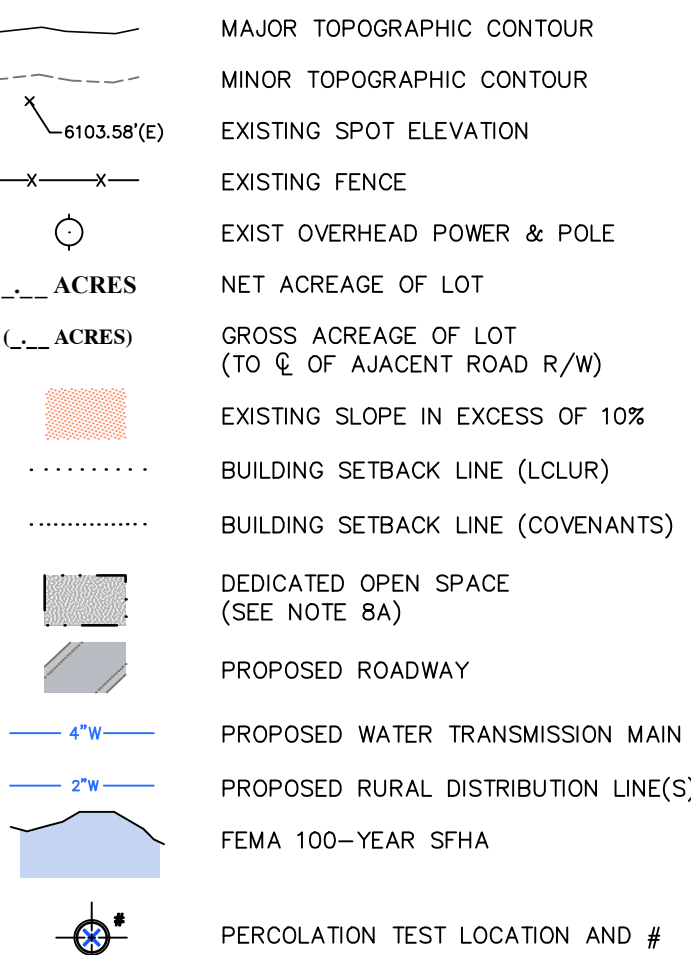
Outside of the State Engineer's Office Control Area

PlanCheyenne- Rural Residential (RR)



This map/data is made possible through the Cheyenne and Laramie County Cooperative GIS (CLCCGIS) Program and is for display purposes only. The CLCCGIS invokes its sovereign and

TABLE (cont.)					
CURVE #	DELTA	RADIUS	CHORD BEARING	CHORD LENGTH	ARC LENGTH
C22	12° 25' 34"	588.46	583° 46' 57"E	82.29	82.45
C23	3° 10' 59"	598.06	578° 40' 24"	33.24	33.25
C24	14° 18' 26"	598.46	567° 55' 42"E	149.05	149.44
C25	2° 13' 41"	598.46	573° 58' 02"E	33.28	33.28
C26	12° 14' 41"	598.46	565° 45' 02"E	125.97	126.15
C27	37° 4' 37"	305.00	542° 34' 34"E	197.56	201.18
C28	2° 36' 40"	305.00	560° 06' 02"E	14.08	14.08
C29	38° 54' 29"	305.00	N39° 22' 27"W	203.16	207.12
C30	23° 40' 47"	305.00	N11° 23' 23"E	125.16	126.05
C31	19° 15' 52"	305.00	N8° 57' 36"E	105.51	106.04
C32	51° 50' 09"	215.00	S25° 55' 16"E	187.95	194.51
C33	28° 09' 34"	335.00	S39° 28' 20"W	162.99	164.64
C34	20° 33' 09"	335.00	N40° 45' 08"E	128.44	129.13
C35	10° 37' 30"	360.00	55° 05' 05"E	66.62	67.42
C36	9° 56' 30"	360.00	55° 13' 08"E	62.34	62.42
C37	6° 13' 43"	360.00	55° 15' 15"E	55.07	55.23
C38	6° 13' 13"	500.00	N61° 15' 41"E	54.26	54.28
C39	44° 24' 34"	120.00	S27° 42' 34"E	90.70	93.02
C40	4° 56' 06" 42"	32	S2° 56' 54"E	10.70	10.71
C41	42° 03' 46"	60.00	N68° 41' 40"E	43.07	44.05



PROPOSED WATER/ SEWER FLOW CALCULATIONS									
Land Use		Contributing Unit Type	Enter No. of Units	Water			Sewer		
				Avg. Flow (gpm)	Max Day Flow (gpm)	Peak Flow (gpm)	Avg. Flow (gpm)	Max Day Flow (gpm)	Peak Flow (gpm)
Residential									
Single Family Homes, Apts., Townhomes, Duplexes	Residence	38		6.3	16.3	33.2	4.9	9.9	27.7

Exhibit A

Date: 3/24/2

Sheet #: 1 of 1



State Engineer's Office

HERSCHLER BUILDING, 2 WEST
CHEYENNE, WYOMING 82002
(307) 777-6150

MARK GORDON
GOVERNOR

BRANDON GEBHART, P.E.
STATE ENGINEER

April 11, 2023

Laramie County Board of Commissioners
309 West 20th Street
Cheyenne, WY 82001

RE: Ironhorse Subdivision, DEQ 2023-064

Dear Commissioners:

This proposed subdivision is to be located in parts of the NE/4 of Section 12, T14N, R67W, Laramie County, Wyoming. The subdivision will be made up of 37 tracts (33 residential) with the average lot size of 2.79 acres. The proposed central water supply is to be provided by two wells in the High Plains Aquifer, with on-lot sewage systems. Based upon the review of both the subdivision application and a preliminary search of the agency's water rights database, the State Engineer's Office offers the following:

The water right search revealed that there are no subject existing water rights of record that attach to the subdivision lands. Given this, further action on the part of the subdivider or his agent in addressing Wyoming Statute 18-5-306 (a) (xi) is not required.

Additional Comments:

1. Our records indicate the planned locations of the two proposed water supply wells and proposed storage tank (see Figure 2 of the Ironhorse Well No.1 Project Design Report, page 2) are in the same location as P67891, Laramie County #10, one of SEO's monitor wells. Therefore, our office stresses that great care must be taken during construction to not disturb the existing monitor well. It is also strongly suggested that the subdivision permit a monitor well on the downgradient of the proposed development to allow the impact of this subdivision's water usage to be studied.
2. If any new wells are proposed, they must be constructed in accordance with the State Engineer's Office Rules and Regulations, Part III, Water Well Minimum Construction Standards. An approved permit from the Wyoming State Engineer's Office is required prior to the drilling of any water well.
3. The proposed subdivision is located within the North Cheyenne Study Area and as such well permits within this area are issued with conditions and limitations regarding well construction requirements.
4. The procurement of the necessary and appropriate State Engineer water right permit allows the applicant to attempt to develop a water supply adequate to meet the proposed needs, and is no guarantee that any water will be physically available.

5. Any well not to be used must be properly plugged and abandoned as outlined in the above referenced rules and regulations.
6. Any wells developed for uses that do NOT fall within the definition of domestic or stock use require adjudication by the Board of Control.

In all correspondence regarding this application, please reference "State Engineer Subdivision Review Number 2023-10-2".

If you have any questions, please feel free to contact me at (307) 777-2975, or if you prefer email, at sue.kinsley@wyo.gov . Thank you for the opportunity to comment on the subdivision application.

Sincerely,



Sue Kinsley
Natural Resources Analyst
TRANSMITTED VIA EMAIL

Cc: Dennis Lewis, P.E., DEQ Water Quality Division, 200 W. 17th Street, Cheyenne, WY 82002
Lisa Lindemann, Administrator, Ground Water Division
Jed Rockweiler, Administrator, Surface Water Division
Cheryl Timm, Administrator, Board of Control Division
Cory Rinehart, Superintendent, Water Division I
Kasey Kiel, Hydrographer/Commissioner, Water Division I, District 1



Department of Environmental Quality

To protect, conserve and enhance the quality of Wyoming's environment for the benefit of current and future generations.



Mark Gordon, Governor

Todd Parfitt, Director

April 20, 2023

Laramie County Board of Commissioners
309 West 20th Street
Cheyenne, WY 82001
Via email: office@laramiecountyclerk.com

RE: **Adverse Recommendation,**
Iron Horse Subdivision, Laramie County
WDEQ/WQD Application #2023-064

Dear Commissioners:

The Wyoming Department of Environmental Quality - Water Quality Division (WDEQ-WQD) has completed the review of the proposed Iron Horse Subdivision located in the Section 12, T14N, R67W, 6th P.M., in Laramie County, Wyoming. The Iron Horse Subdivision will create 33 lots out of approximately 82.6 acres. The information was submitted by Matthew Kaufman, Manager, and was signed by John Wetstein, PE, PG, a Wyoming Licensed Engineer and Wyoming Professional Geologist. This letter represents the recommendations of the Wyoming Department of Environmental Quality as required by W.S. 18-5-306(c).

The subdivision review consisted of (i) review of the proposed sewage system, (ii) review of the proposed water system, and (iii) groundwater review of the proposed sewage system. The conclusion of WDEQ-WQD's subdivision review results in either a "Non-Adverse" or "Adverse" recommendation.

- **Non-Adverse** - - the proposed subdivision meets the safety and adequacy requirements of WDEQ Chapter 23 Rules
- **Adverse** - - the proposed subdivision does not meet the safety and adequacy requirements of WDEQ Chapter 23 Rules
- **Notes Required to be Added to the Final Plat** - - an Adverse recommendation will result in comments to address water or wastewater concerns with specific "Notes Required to be Added to the Final Plat."

PROPOSED SEWAGE SYSTEM REVIEW:

The proposed sewage system consists of individual on-site septic tanks and leach fields, constructed to standards established in WDEQ Water Quality Rules (WQR) and standards established by the Cheyenne/Laramie County Environmental Health Department.

- The information submitted addresses the safety and adequacy of the proposed sewage system.
- The information submitted addresses the safety and adequacy of the proposed sewage system, provided the statements indicated in the "Notes Required to be Added to the Final Plat" are adhered to.

PROPOSED WATER SYSTEM REVIEW:

The subdivision's proposed water system consists of well(s) to be operated by the Ironhorse Improvement and

200 West 17th Street, Cheyenne, WY 82002 · <http://deq.wyoming.gov> · Fax (307)635-1784

ADMIN/OUTREACH (307) 777-7937	ABANDONED MINES (307) 777-6145	AIR QUALITY (307) 777-7391	INDUSTRIAL SITING (307) 777-7369	LAND QUALITY (307) 777-7756	SOLID & HAZ. WASTE (307) 777-7752	WATER QUALITY (307) 777-7781
----------------------------------	-----------------------------------	-------------------------------	-------------------------------------	--------------------------------	--------------------------------------	---------------------------------

Service District (I&SD). Each well must be permitted by and constructed to the WQR Chapter 26 Well Construction Standards and the Wyoming State Engineer's Office standards.

- The information submitted addresses the safety and adequacy of the proposed water system.

PROPOSED SEWAGE SYSTEM GROUNDWATER REVIEW:

The proposed sewage system consists of individual on-site septic tanks and leach fields constructed to standards established in WQR and standards established by the Cheyenne/Laramie County Environmental Health Department. The sub-division does overlie an unconfined Class I groundwater as defined by WQR Chapter 8 Rules.

- The information submitted addresses the safety and adequacy of the proposed sewage system.
- The information submitted addresses the safety and adequacy of the proposed sewage system, provided the statements indicated in the "**Notes required to be added to the final plat**" are adhered to.

Per WQR Chapter 23 Section 7(d)(viii)(A)(I), WDEQ requires that subdivision applications demonstrate that nitrates from on-lot sewage systems will not exceed 10 mg/L at the downgradient property boundary. WDEQ recommends including a requirement on the plat that all septic systems be enhanced treatment systems that include advanced nitrogen removal technology. Per WQR Chapters 11 and 23, all enhanced treatment systems must be permitted and approved by the WDEQ. A Wyoming Registered Professional Engineer must design the enhanced treatment systems (WQR Chapter 23, Section 7(d)(viii)(E)).

Please refer to Appendix A, Groundwater Loading Model analysis located at the end of this letter for additional information.

NOTES REQUIRED TO BE ADDED TO THE FINAL PLAT:

In order to meet requirements established in Wyoming Water Quality Rules, Chapter 23 Section 7(d)(viii)(A)(I), individual septic systems are required to be enhanced treatment systems with advanced nitrogen removal technology that is capable of achieving a nitrate concentration of less than 10 mg/L at the downgradient property boundary.

A Wyoming Registered Professional Engineer must design the enhanced treatment septic system per WQR Chapter 23, Section 7(d)(viii)(E). A permit to construct application for enhanced treatment septic systems must be submitted to the Southeast District Engineer in Cheyenne. For the proposed Advanced Enviro-Septic enhanced treatment septic systems the bottom of the leach field bed sand must be at least 6 feet above seasonally high groundwater.

CONCLUSIONS:

Based on the materials and information presently submitted for this Chapter 23 subdivision application, the WDEQ is issuing an "**Adverse**" recommendation applicable to the Iron Horse Subdivision. The adverse recommendation is due to the determination that nitrate levels will exceed 10 mg/L at the downgradient property boundary if enhanced treatment systems with advanced nitrogen removal are not used. The WDEQ recommends to the County Commissioners that the subdivision be approved as presently submitted using enhanced wastewater treatment systems. The WDEQ recommends including the **Notes Required to Be Added to the Final Plat** and proposed wastewater systems modified to achieve the standards within WQR Chapter 23 Rules.

- The subdivision needs to address nitrate levels exceeding 10 mg/L at the downgradient property boundary in the analysis through the use of enhanced treatment systems with advanced nitrogen removal technology.

Disclaimer:

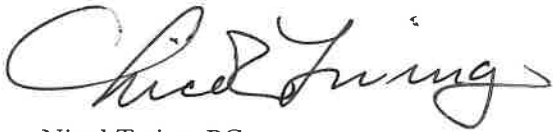
The Adverse recommendation provided above has nothing to do with, nor has any bearing on the water rights associated with the Iron Horse Subdivision. Any questions or concerns about the water rights for the Iron Horse Subdivision should be directed to the State Engineer's Office.

Nothing in Department of Environmental Quality recommendations or comments regarding the proposal for the Iron Horse Subdivision shall be construed to relieve Matthew Kaufman, or Ironhorse Capital, LP, of the obligation to obtain any permits or additional approval from any local, state or federal agencies as required by law, rules, regulations, or ordinances. Nothing in these recommendations commits the Department of Environmental Quality or Laramie County to the issuance of required permits for construction, operation, or modification of water supply or sewage systems.

Sincerely,



Dennis Lewis, PE
South East District Engineer
Water & Wastewater Section, Water Quality Division



Nicol Twing, PG
WDEQ Groundwater Geology Supervisor
Groundwater Section, Water Quality Division

Emailed copies:

Matthew Kaufman, Manager Ironhorse Capital, LP, 2525 Warren Ave. Suite 500,
mkaufman@hkwwyolaw.com

John Wetstein, PE, PG, Engineering Associates, PO Box 2202, Laramie, WY 82073,
john.wetstein@eaengineers.com

Appendix A
Ground Water Loading Analysis

Wastewater System Discharges to same aquifer that supplies a public water supply and are located within Zone 1 or Zone 2

- Determination = No

Chapter 23 Appendix A – Nitrate Loading Analysis

Nitrate Loading – Co

Number of Bedrooms	Consultant Analysis	DEQ Analysis
4 Bedrooms	14.1 mg/l	14.1 mg/l
5 Bedrooms		14.28 mg/l
6 Bedrooms		15.24 mg/l

Chapter 23 Appendix B – Minimum Isolation Distances Analysis

Vertical Travel Time – t_1

Number of Bedrooms	Consultant Analysis	DEQ Analysis
4 Bedrooms	1,021 days	1,021 days

Horizontal Travel Time – t_2

Number of Bedrooms	Consultant Analysis	DEQ Analysis
4 Bedrooms	88 Feet	100 Feet defaulted

Chapter 23 Appendix C – Public Water Supply Analysis

No Changes were made to the analysis

Applicant Response to 2nd Round Review Comments 12/07/22

ATTACHMENTS:

- A. Iron Horse Subdivision WDEQ Chapter 23 Review Report (9/14/2022) by EA Engineers
- B. Non-Adverse Recommendation (WDEQ/WQD Application #2022-300)
- C. Approved Resolution (221206-17) Establishing the Ironhorse Improvement & Service District (IISD)
- D. Wyoming Water Development Commission Letter of Concurrence
- E. Capacity Memo from City of Cheyenne Board of Public Utilities
- F. Ironhorse Preliminary Water System Design Report by Summit Engineering

Cheyenne Development Services: Cheyenne Development Services, Seth Lloyd
Comments Attached 11/01/2022

- 1. Original official comment letter not adequately resolved. Letter remains 'as is'.

✓ **Acknowledged.**

- 2. Plan for alternate, non-BOPU water source (if necessary) understood.

✓ **Acknowledged.**

Cheyenne MPO: Cheyenne MPO, Christopher Yaney Comments Attached
10/26/2022

- 1. ERIS - Page 2 of 3, The information under "Transportation" still does not match Trip Generation by Use on the PDP map dated 10/25/2022. There are two different rates being used. Please correct to match ITE Trip Generation 11th Edition on both ERIS and PDP map dated 10/25/2022. Rates are not correct.

**Removed from ESIR to avoid confusion and eliminate discrepancies.
Corrected on Revised PDP.**

- 2. Currently there is 40' of right-of-way Buck Brush Rd running from Ole Maverick Rd to W Riding Club Rd. Buck Brush Rd is spaced approximately 1/4 mile between Hynds Blvd and Yellowstone Rd. This would provide connection to the north as future land is developed. Therefore, this development shall provide the additional 40' right-of-way for a total of 80' right-of-way along the eastern side of this development. In areas that the road alignment is shifted, this development shall continue to provide the 80' right-of-way and connect to the new Ole Maverick Rd within this development. This will provide a future second roadway for surrounding land to develop. The current plat listed for Sunset Tracts-Third Filing shows 40' right-of-way and needs to remain for future north south connections. There is no document that shows this 40' right-of-way and been changed to a 40' Non-Motorized Public Access Easement. If so, please provide the document with date showing changes.

Due to the floodplain and intervening developed properties, it is highly unlikely Buckbrush road will ever be developed to a full R/W width. Following City Comment and pursuant to the Laramie County Comprehensive Plan (LCCP) Policy 4.1.E, the existing 40' along the property boundary is retained on the Revised PDP map as a "40' NON-MOTORIZED PUBLIC ACCESS EASEMENT" connecting to the existing 40' R/W north of the project. See updated PDP map dated 12/07/22.

- 3. Since the PDP map dated 10/25/2022, the Traffic Impact Study needs to be updated to reflect the trip generated by the RV/Boat Storage site within this development.

Updated TIS (November 2022) submitted with Revised PDP.

4. Again, this development should not be allowed city water to reduce lot size without annexing into the city. Therefore, the development shall only build out to Laramie County minimum lot size. If the development is concerned about well water contamination, then the development needs to determine what size lots need to be to prevent water contamination.

See attached Iron Horse Subdivision WDEQ Chapter 23 Review Report (9/14/2022) by EA Engineers and WDEQ Non-Adverse Recommendation (WDEQ/WQD Application #2022-300)

Cheyenne Planning: Cheyenne Development Services, Seth Lloyd No Comments 11/01/2022
No Comments

County Engineer: County Engineer, Scott Larson Comments Attached 10/28/2022 All previous comments have been adequately addressed. Therefore I have no further comments at this time.

County Public Works Department: County Public Works Department, Molly Bennett No Comments 11/07/2022
No Comments

County Real Estate Office: County Real Estate Office, Laura Pate No Comments 10/31/2022
No Comments

Environmental Health: Environmental Health Department, Tiffany Gaertner Comments Attached 11/02/2022
Regulations: Laramie County Small Wastewater System Regulations

Comments:

Five acres are required for lots utilizing private wells. At least 2.5 acres are required for lots utilizing a public water supply. Most lots in Iron Horse 3rd filing do not meet the minimum lot size requirements.

Proposed minimum lot sizes & request for Lot Size Waiver based on conversations with Environmental Health dating back to July 2020 (with former director) and supported by the IISD Gross Lot Acreage (2.79-ac gross/2.0-ac net across 3 Filings of Ironhorse); health & safety is furthermore reinforced by the DEQ Chapter 23 Report and WDEQ Recommendations. As noted in the non-adverse recommendation: *if the proposed BOPU water service is not approved: “the use of on-site wells will need to be evaluated for Nitrate levels”.*

Septic systems must be kept fifty (50) ft from all property lines.

Septic setbacks were not shown on submitted PDP for clarity’s sake. Please see attached Revised PDP 50’ setbacks turned on for all residential lot lines. (Sporadically annotated for reference).

Leach fields on PDP map are not at scale and are not 50 ft from all property lines.

Applicant Response to 2nd Round Review Comments 12/07/22

All leach fields on the PDP map were, and are, shown to scale at 1,140 sf ea. (2 fields @ 38'x30' [2,280 total]) – this is slightly larger than required by §7(d)(v) of the attached Iron Horse Subdivision DEQ Chapter 23 Review Report:

“Using this soil-loading rate and an effluent discharge of 470 gallons per day (assume four-bedroom home with design flow of 117.5 gallons/bedroom/day); the required area for the leach field is approximately 1,022 ft². The area required for the leach field represents approximately 2.35% of an acre. To account for a replacement leach field, approximately 2,044 square feet will need to be set aside.”

We have dimensioned the primary & replacement fields on LOT 6 of Revised PDP.

Setback from public water supply well to septic systems is 200 ft,

If project necessarily needs to pivot to public supply well: 200' radius shown from “80'x80' Water Storage & Well Easement” (if the easement needs to accommodate a well, it may need to expand to the east slightly (i.e. 110'x80')).

If on-site supply wells are used, a new Chapter 23 Application will need to be submitted to DEQ. This will be finalized before final plat application.

and buried water lines require a 25 ft setback from septic systems.

✓ **Acknowledged.**

Septic systems are not allowed under irrigated landscaping or lawns.

✓ **Acknowledged. “Typical Lot Development Footprints” are schematic and provided solely for scale & context.**

A variance is required for lots smaller than 2.5 acres utilizing a public water supply. Lots smaller than 2.0 acres have difficulty meeting the minimum required setback distances are not recommended.

✓ **Acknowledged. Per Environmental Health Direction August 16, 2022:**

1.25 acres (net) would be the minimum acceptable lot size for Environmental Health support of waiver request (gross minimum acreage of 2.5-acres required).

DEQ is requesting that enhanced systems are used to reduce nitrates for homes with five or more bedrooms.

Developer intends to require enhanced septic systems on all lots.

Provide copy of Chapter 23 study to Environmental Health Office.

Chapter 23 Study & Report and non-adverse recommendation was provided to County Planning Department with applications. Also attached to this re-submittal.

Signed final plat must be submitted to Environmental Health Office prior to application for small wastewater permits.

✓ **Acknowledged. If this Preliminary Development Plan (PDP) is approved, Owner/Developer will proceed with appropriate diligence, feasibility, and development applications. Opportunity for more detailed review will be afforded at time of final plat.**

Intraoffice: Planners, Cambia McCollom No Comments
10/31/2022 No Comments

Laramie County Fire Authority: Laramie County Fire Authority, Manuel Muzquiz
No Response 11/14/2022
No Comments

Planners: Planners, Mason Schuricht No Response
11/14/2022 No Comments

WYDOT: WYDOT, Taylor McCort Comments Attached 10/31/2022
No additional comments, see prior comments.

WYDOT: WYDOT, Taylor McCort Comments Attached 09/26/2022

1. Developers should be aware that any work or presence in the right of way created by development/construction for this project will need the appropriate permitting or licensing between the utility owner (or appropriate local agency for fence modifications) and WYDOT District 1 Maintenance. Also, the development must maintain historic drainage corridors so that drainage is not diverted to other entry points to the R/W. If drainage is affected in the highway right-of-way, a drainage study needs to demonstrate that post-development discharge rates are metered at or below pre-development rates for 2, 5, 10, 25, 50 and 100 year events and will need to be reviewed by WYDOT Bridge/Hydraulics Program. Please contact Michael Elliot (Michael.Elliott@wyo.gov -- 307.745.2123) for utility permit information.

✓ **Acknowledged**

2. Access roads from the state highway system will require WYDOT access permits for new/modified/change in use, including removal. Maintain existing drainage patterns so that stormwater is not concentrated and diverted from locations where it currently crosses WYDOT R/W. Please contact Mike Ginther (Michael.Ginther@wyo.gov- 307.745.2118) for access permit information.

✓ **Acknowledged**

3. The accesses along Hynds Blvd. appear to meet WYDOT spacing requirements. The access on Yellowstone Rd. does not appear to meet spacing requirements. This comment does not count as an official evaluation. All spacing requirements will be officially determined when access applications have been submitted.

Existing Residential Access at Yellowstone Road was reviewed with Ironhorse, 1st Filing with the proposed re-alignment of West Ole Maverick Road, which significantly improves safety from the current mis-alignment. Access Permit Applications will be submitted to WYDOT at the time of Final Plat or before, and as noted in the TIS, spacing may require an exception from the Access Review Board.

4/17/23

Subject: Comments on Ironhorse Preliminary Development Plan

Planning Commission/Board of Commissioners:

I own property adjacent to the proposed Ironhorse subdivision. I will not be able to attend the May 11, 2023 public hearing on this PDP. My comments follow.

The Preliminary Development Plan & Land Analysis Map attached to my "Dear Adjacent Landowner" letter does not appear to match the various documents in the "Citizen Access – Project or Permit Information" on-line file.

1. This map does not illustrate a "Lot 2", does that lot no longer exist?
2. Note 6(I) on the map states that lots 4 thru 19 are going to have individual septic systems. What about Lots 1-3? WDEQ/WQD regulations require at least 2.5 acres for lots utilizing a public water system. As per the lot acreage displayed on the map, Lots 12-17 do not meet that 2.5 acre minimum lot size and Lots 20-34 do not meet this 2.5 acre minimum lot size.
3. It does not appear that this most recent filing is requesting a variance to this rule? I am against any variance to the minimum lot size.
4. Note 6(II) on the map states that lots 20-36 are going to be tied into an enhanced community septic field. I can find no documents in the project portal that provide any information on the design or operation and maintenance requirements for this system. I would like to see this documentation, prior to the public comment period.

In summary, the adjacent and surrounding land use is primarily comprised of Agricultural Residential developments on 5 acres lots. This is the minimum required lot size for lots with wells and septic systems. The minimum required lot size for lots with a public water system and on-site septic is 2.5 acres. The Ironhorse Preliminary Development Plan continues to propose 33 residential lots with only 10 lots meeting the required minimum lot size of 2.5 acres (assuming the City allows them to use City water).

- There are NO unusual physical circumstances or conditions that necessitate this small of a lot size.
- Strict enforcement of the regulation would NOT be detrimental to the existing property, adjacent property, or the existing environment.
- The property CAN BE reasonably developed in a manner substantially equivalent to the other landowners in the neighborhood.
- Allowing septic systems on lot sizes smaller than 2.5 acres has substantial potential to harm the public safety and welfare of neighboring drinking water.
- These small lots WILL alter the essential character of the neighborhood or district.
- These small lots WILL substantially or permanently impair the reasonable use and enjoyment or development of adjacent property.

Thank you,

Scott & Ruth Benson
409 West Riding Club Rd.



College of Business
Center for Business
and Economic Analysis

Evaluating the Economic Impact of Providing Municipal Water to Laramie County Housing Developments

April 2023

Center for Business and Economic Analysis

The Center for Business and Economic Analysis (CBEA) at the University of Wyoming (UWyo) supports the economic growth and diversification of Wyoming's economy through applied economic and business analytics for communities, industries, and entrepreneurs. The center was established in 2019 as a unit within the College of Business. CBEA is a member of the Association for University Business and Economic Research (AUBER).

Team

Dr. David Aadland is the Director of the CBEA at UW. He is a professor and former chair of the Department of Economics. He teaches Macroeconomics and Econometrics. His research interests have been wide and varied over the years. He has written papers in the areas of macroeconomics, non-market valuation, applied econometrics, behavioral economics, agricultural economics, and statistics. His recent research has focused on interdisciplinary natural resource questions and the intersection of ecology, epidemiology, entomology, and economics. In particular, he has been working on projects related to pine beetle epidemics, ecological impacts of energy development, and the economics of chronic wasting disease. He holds a Ph.D. in Economics from the University of Oregon.

Dr. Anne Alexander served as Interim Provost and Vice President of Academic Affairs at the University of Wyoming. She earned her Ph.D. in Economics at UW, and her Master's and BBA in Economics from New Mexico State University. Dr. Alexander's research and teaching focuses on the history of American capitalism, international economics, and health economics. She regularly speaks to statewide audiences and state and national media about the status of the Wyoming, US, and global economies. She produces an annual economic and statewide engagement report on the impacts of the University on the state, and she has authored several publications on the political and economic impacts of transboundary pandemics and the productivity effects of various diseases. Prior to her position in Academic Affairs, she was the Director of International Programs and Associate Dean of Outreach at UW. She has also served as a Resources Economist in the U.S. Department of State, Assistant Dean of the UW College of Business, and Director of the interdisciplinary Health Economics Policy Center at UW. She is an alumni of Leadership Wyoming, Class of 2013.

Dr. Alex Gebben is an Energy Economist conducting research with the Center for Business and Economic Analysis. He holds a B.S. in Petroleum Engineering and graduate degrees in Mineral and Energy Economics from the Colorado School of Mines. This background has led him to diverse research topics in the areas of ground water management, oil industry incentives, and unconventional methods of lithium recovery. He is particularly interested in how the industries of oil and gas, wind, geothermal, and nuclear power embody the responsive relationships between policy, natural resources, and economic outcomes. Before joining the CBEA he conducted research projects in conjunction with the Critical Materials Institute, as well as the USDA. Alex is an avid fly fisherman with a love for the outdoors, as well as a self-taught programmer.

Dr. Rob Godby is an associate professor in the Economics Department at UW, former Associate Dean of the Haub School of Environment and Natural Resources, and former interim Dean of the College of Business. He also served as the Deputy Director of the University of Wyoming's Center for Energy Regulation and Policy (CERP). Additionally, he serves as a Daniel's Fund Ethics Initiative Faculty Fellow and is an adjunct faculty member with the MBA program at Pforzheim University in Germany. Outside his academic duties, Rob was appointed to serve on the State of Wyoming's Consensus Revenue Estimating Group in 2019. His research areas include natural resource, energy and environmental economics, industrial organization, and macroeconomic policy. He is often interviewed by national and international news media on energy and macroeconomic issues. Outside of work, Rob enjoys spending time being walked by his dogs. His other passions include sports-car and bicycle racing, both of which he has participated in rather unsuccessfully.

Dr. Morgan Holland graduated with a PhD in economics from Florida State University in 2022 with research interests in the economics of automation, corporate finance, human capital, and the economics of disability. In addition, Morgan worked as an economic consultant for the FSU Center for Economic Forecasting and Analysis, pursuing applied research projects in a wide variety of fields. Today, Morgan focuses on economic research in the tourism and hospitality sectors in Wyoming. He is especially interested in researching and promoting policies that will help Wyoming take full advantage of its outdoor recreation resources while preserving them for future generations. Morgan also provides support services to stakeholders in the tourism and hospitality sector through economic research and analysis.

Citation: Aadland, D., A. Van Sandt, and S. Nevels. (2023). *Evaluating the Economic Impact of Providing Municipal Water to Laramie County Housing Developments*. Laramie, WY: Center for Business and Economic Analysis (CBEA). Prepared for Cheyenne Mayor's Office.

Correspondence: For additional information, contact aadland@uwyo.edu.

Table of Contents

Executive Summary	5
1. Introduction	6
2. Costs of Providing Municipal Water	8
3. Benefits of Providing Municipal Water	11
4. Spreadsheet Tool	13
5. Assumptions, Limitations and Recommendations	15

Executive Summary

The University of Wyoming Center for Business and Economic Analysis (CBEA) is conducting an economic analysis of providing municipal water to housing developments in Laramie County.

The analysis shows that ...

- for nearly all the parameter values, it does not make sense to provide municipal water to Laramie County new housing developments.
- the loss to the city of Cheyenne for a five-acre new housing development is **\$14,328** or **\$626** per new County resident, using our preferred parameter assumptions.
- the net benefits (i.e., benefits less costs) are negative for all the parameter combinations using the interactive spreadsheet tool.

1. Introduction

In this project, we calculate the economic costs and benefits of providing municipal water to new housing developments in Laramie County, but adjacent to the city of Cheyenne.¹ The analysis is done from the perspective of the city of Cheyenne and only considers items that lead directly to an increase or decrease in the city's budget.

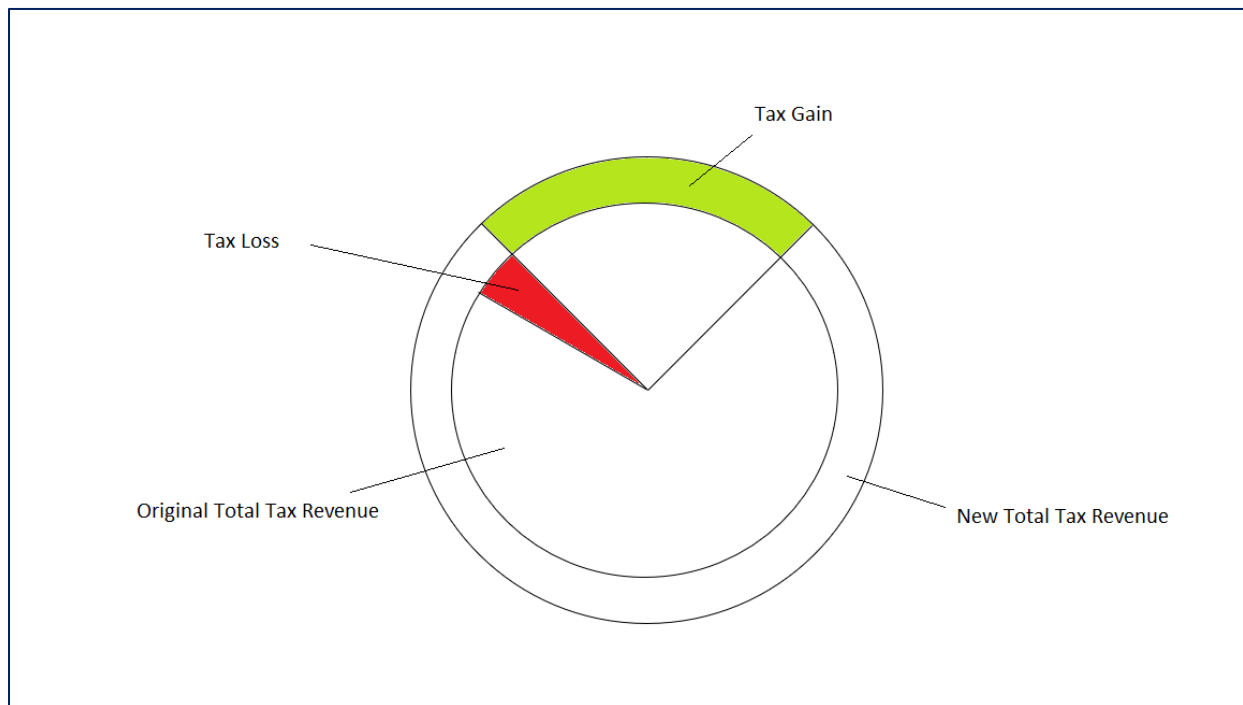
The costs of providing municipal water are items that either require direct expenditures or lead to tax losses for the city of Cheyenne. The tax losses are associated with a reduction in the Cheyenne population when households move from the city to the new housing developments in the county. The reduction in the Cheyenne population relative to the Laramie County population reduce the share of tax revenues allocated to the city of Cheyenne. The items in the analysis include: (1) loss of 4% sales tax revenue, (2) loss of 1% sales tax revenue, (3) loss in gas tax revenue, (4) loss in severance tax revenue, (5) loss in direct distribution, and (6) reduction in the quality of city services.

The benefits of providing municipal water are items that generate more tax revenue for the city of Cheyenne. The items include: (1) one-time 4% sales tax gain from new construction, (2) recurring 4% sales tax gain from new residents, (3) one-time 1% sales tax gain from new construction, (4) recurring 1% sales tax gain from new residents, and (5) gain in gas tax revenue from new residents.

The tax gains and losses are illustrated in Figure 1 below. The two circles represent the total taxes collected, before the new housing development (inner circle) and after the new housing development (outer circle). The slices of the circle represent the portion of tax revenue allocated to the city of Cheyenne. While the new housing development with municipal water increases the total tax revenues as new households move into Laramie County, it also decreases the share allocated to the city of Cheyenne as people move out of the city to the new housing development. The green shaded area represents the gain in taxes for the city of Cheyenne (i.e., more total tax revenues but at a smaller proportion). The red shaded area represents the loss in taxes for the city of Cheyenne (i.e., a smaller proportion of the original tax revenues).

¹ Click [here](#) to see a development map of Cheyenne and the surrounding area.

Figure 1. Tax Gains and Losses to the City of Cheyenne from a New Housing Development with Municipal Water



Sections 2 and 3 below describe the costs and benefits more thoroughly. Section 4 describes the interactive spreadsheet tool, which calculates the net benefits of providing municipal water. The spreadsheet allows the user to vary some key parameters and shows the net benefits to the city of Cheyenne for two cases: (1) the status quo without municipal water that requires one new house per five acres; and (2) providing municipal water to new housing developments in Laramie County. Section 5 discusses the assumptions, limitations and recommendations associated with the analysis.

2. Costs of Providing Municipal Water

We consider eight different economic costs to the city of Cheyenne if they provide municipal water to new housing developments in Laramie County. We list the economic costs and our estimate below. Each item is then discussed in turn.

Economic costs of providing municipal water:

1) *Expansion of municipal water infrastructure.*

Estimate: \$0.

Providing municipal water to housing developments outside the city limits requires expenditures on water infrastructure. We assume these costs are passed along to the housing developers and then to the households purchasing the houses. Therefore, we assume the costs to the city are zero.

2) *Stress on the municipal water system.*

Estimate: \$0.

New housing developments outside the city will also require additional water and more water treatment. Based on interviews with the Cheyenne Board of Public Utilities (BOPU), there is sufficient capacity right now to add more homes and any costs incurred by the city of Cheyenne or BOPU are likely to be compensated for the new homes by charging county residents 1.5 times the water rate as city residents. Therefore, we assume the costs to the city are zero.

3) *Stress on city services.*

Estimate: \$14,355.

Just because someone lives just outside of city limits does not mean they cease to use the city's services. These county residents are still likely to recreate in Cheyenne, use police and fire services, and the city's roads. Indeed, many of these county households likely conduct their daily economic activity in the city limits of Cheyenne. Unfortunately, Cheyenne does not benefit from these residents living just beyond city limits from a tax standpoint. Thus, the city of Cheyenne would incur a cost if it sought to maintain the same per capita public expenditures as more county residents use its services. To estimate Cheyenne's cost of providing services to county residents we first calculated the per household public expenditures by totaling categories in the city's 2022 adopted budget that are reasonably tied to population. For example, within the Mayor Department, we may not expect the costs to the mayor division to increase with more county residents, but we would expect higher costs to the animal control, compliance, municipal court, and youth alternatives divisions. We estimate the status quo

expenditures per household to be \$1,466. As more out of county residents are added, this figure becomes a per new-county-household cost in order to maintain the same level of services.

4) *Decrease in 4% state sales tax revenues.*

Estimate: \$6,945.

If municipal water is provided to housing developments in Laramie County, the portion of the 4% sales tax revenue allocated to the city of Cheyenne is likely to fall. Laramie County receives 30% of the 4% sales tax revenue and 70% goes to the state of Wyoming. In 2022, the 30% share amounted to \$37M. The 30% is then divided between five cities/county: Cheyenne, Albin, Burns, Pine Bluffs, and Unincorporated Laramie County, as a proportion of their respective populations. Therefore, as households move from the city to the new development in the county, the city of Cheyenne receives a smaller portion of the 4% sales tax (i.e., the red shaded area in Figure 1 above).

5) *Decrease in 1% city sales tax revenues.*

Estimate: \$5,631.

Similar to the tax loss in item #4 above, the city of Cheyenne loses a portion of the 1% sales tax revenue. In 2022, the 1% sales tax generated \$30M for Laramie County without any portion going to the state. The \$30M is then divided in the same fashion described in item #4 above.

6) *Decrease in gas tax revenues.*

Estimate: \$10.

The city of Cheyenne also loses gas tax revenues if households residing in Cheyenne move to the new housing development in Laramie County. However, the allocation rule for the gas tax is different than sales taxes. 15% of the gas tax revenue is allocated to municipalities with Cheyenne's portion weighted by 75% from gas tax distributors in Cheyenne and 25% from Cheyenne's portion of the total incorporated population of Wyoming. In 2022, the total allocation was \$1.59M. The loss in taxes associated with the new housing development is due to having a smaller portion of Wyoming's incorporated population as households move from Cheyenne to unincorporated Laramie County (i.e., see shaded red area in Figure 1).

7) *Decrease in severance tax revenues.*

Estimate: \$53.

The city of Cheyenne received \$2.2M in severance taxes from the state of Wyoming in 2022. The total allocation to all incorporated cities was \$15.656M in 2022. If a new five-acre housing development with municipal water is created in Laramie County and half of the houses purchased are from Cheyenne residents, then the loss in severance taxes to the city of Cheyenne would be \$53. This is illustrated by the shaded red area in Figure 1.

8) *Decrease in direct distribution from the state of Wyoming.*

Estimate: \$833.

The direct distribution of tax revenue from the state of Wyoming to the city of Cheyenne in 2022 was approximately \$4.8M. The allocation mechanism uses the Madden-Bebout rule. The Madden-Bebout rule specifies that 75% of the distribution is from sales tax per capita and 25% of the distribution is from assessed valuation per capita plus a fixed amount of \$35,000. If a new five-acre housing development is created and half of the houses are purchased by Cheyenne residents, then the direct distribution to Cheyenne would decrease by \$833.

Total Economic Cost to the City of Cheyenne: \$27,808

3. Benefits of Providing Municipal Water

We consider five different economic benefits to the city of Cheyenne if they provide municipal water to new housing developments in Laramie County. We list the economic benefits below and discuss each in turn.

Economic benefits of providing municipal water:

- 1) *One-time 4% sales tax increase from new house construction.*

Estimate: \$4,535.

When new homes are built, they create local economic opportunities for both labor and suppliers of input materials. While the city and county do not encounter direct tax revenue from the labor, they do collect on the sales tax of the input material suppliers. We estimate the value of these input materials by modeling a new home construction in the economic contribution software, IMPLAN. First, we assume that an average new home in Cheyenne costs \$267,000 to build. This estimate was calculated by cross-referencing the average construction cost per square foot in Cheyenne and multiplying it by the average home size in Cheyenne. Given this initial construction cost, IMPLAN estimates the total value added to Laramie County's economy by calculating all purchases not only from the home's construction, but also the purchases of new labor used to build the home. In total, IMPLAN estimates a value added contribution of \$464,029.21 from new home construction in Laramie County. Multiplying this figure by the 4% sales tax and Cheyenne's new (lower) tax distribution results in Cheyenne's benefit from the construction of a new home in the county. Since a new home is only built once, this economic contribution only takes place in the year the home is built.

- 2) *Recurring 4% sales tax increase from new county residents.*

Estimate: \$2,731.

While more residents living in Laramie County relative to the city of Cheyenne would lead to a sales tax distribution that favors the county more, total sales tax revenue could increase due to new residents moving to the county. In Figure 1, this concept is represented by the green area. Unlike new home construction, this tax revenue would be a recurring source of revenue for the city and county since the overall population of the county and related sales activity would likely increase and stay at these new higher levels, assuming no additional shocks. This means that there is potential for the city of Cheyenne to benefit from new home construction in the county if the negative distributional effect (red area in Figure 1) is counteracted by more out-of-county people moving into the new homes (green area in Figure 1).

3) *One-time 1% sales tax increase from new house construction.*

Estimate: \$3,779.

The one-time 1% increase in sales tax revenue from new house construction is similar to the benefits in item #1 above. However, all the revenue from the penny tax goes to Laramie County.

4) *Recurring 1% sales tax increase from new county residents.*

Estimate: \$2,214.

The recurring 1% increase in sales tax revenue from new county residents is similar to the benefits in item #2 above. However, all the revenue from the penny tax goes to Laramie County.

5) *Gas tax increase due to new residents.*

Estimate: \$221.

Since IMPLAN takes into account input suppliers' purchases of gas, the only additional increases in gas tax revenue come from new residents' purchases of gas. This means that this tax benefit is incurred every year the new residents live in the county. We calculate this by multiplying the per capita gas tax revenue by the number of new residents in the county.

Total Economic Benefit to the City of Cheyenne: \$13,479

Table 1. Economic Costs and Benefits of Providing Municipal Water

	Total (\$)	Total per Person (\$)
Costs	\$27,808	\$1,214
Benefits	\$13,479	\$589
Net Benefits	(\$14,328)	(\$626)

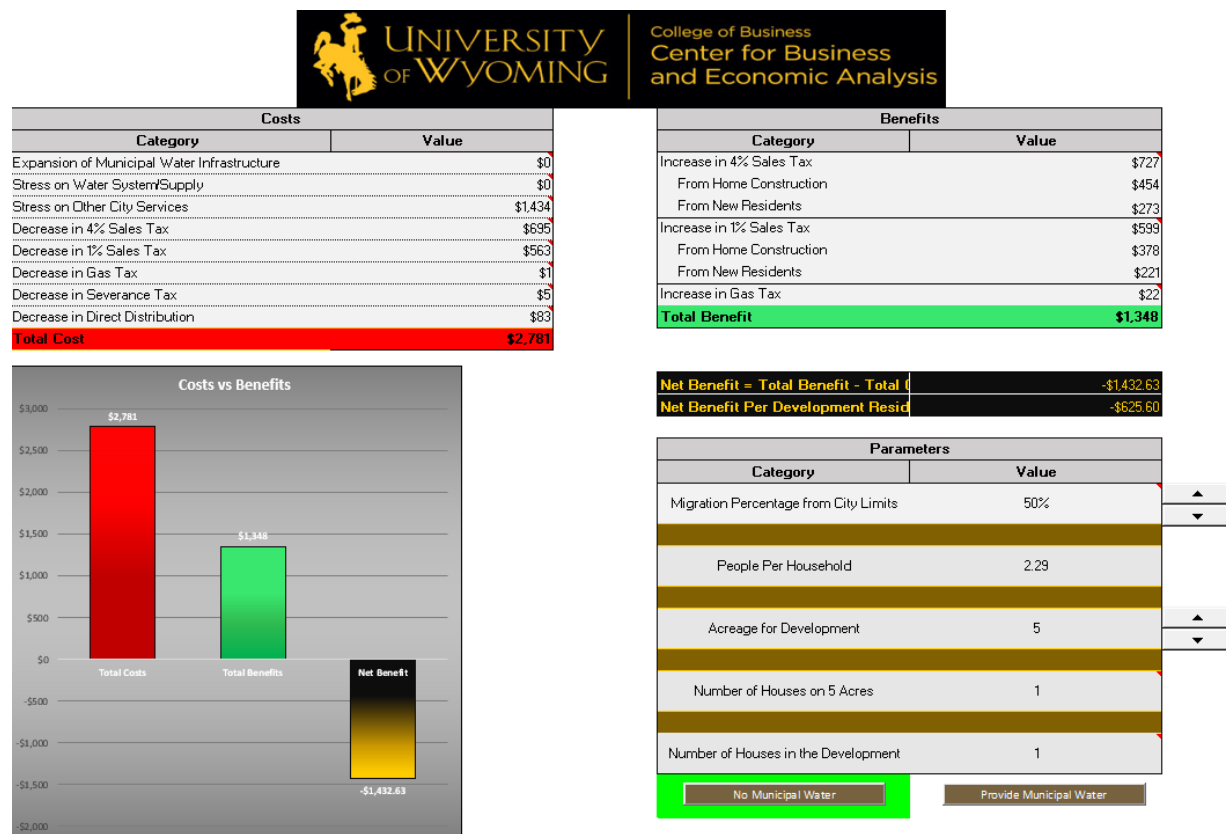
4. Spreadsheet Tool

We also provide an interactive spreadsheet tool which allows the user to see the economic costs and benefits. The tool allows the user to vary key parameters to see how the net benefits (i.e., economic benefits less costs) change as the parameter values are changed.

Figure 2. Interactive Spreadsheet Tool

We also provide an interactive spreadsheet tool which allows the user to see the economic costs and benefits. The tool allows the user to vary key parameters to see how the net benefits (i.e., economic benefits less costs) change as the parameter values are changed.

Figure 2. Interactive Spreadsheet Tool



Above is a picture of the layout of the interactive spreadsheet. The purpose of the tool is to allow for easy visualization of the results of the analysis and for varying the parameters for different conditions.

The spreadsheet is divided into four primary regions. In the top left are the costs we take into account and their values based on the parameter levels set in the tool. In the top right are the benefits derived from the chosen parameter levels. In the bottom right are the parameters used in calculating the aforementioned costs and benefits. Two of the parameters we hold fixed; the

number of people per household and the number of houses on 5 acres. The value of the people per household parameter comes from the US Census' data on the city of Cheyenne. The number of houses on 5 acres, if municipal water is provided, we assume to be 10. This accounts for roads and other infrastructure in the development that is not part of individual lots. Two parameters are directly variable; the migration percentage from the city and the number of acres being considered for development. The migration percentage affects the values of the costs and benefits emanating from taxation by changing the population of Cheyenne relative to the county. A value of 100% indicates that all residents of the new homes have emigrated from Cheyenne, while a value of 0% indicates that all the residents are from outside the area. At the bottom of the parameters table are two buttons that set the plan being considered in the tool: either no provision or municipal water or provision of municipal water. Clicking between these varies the number of houses on 5 acres and resets all the parameter values to base assumptions. Finally, in the bottom left is an interactive graph showing the sizes of the costs and benefits relative to each other and the net benefit per resident of the new homes.

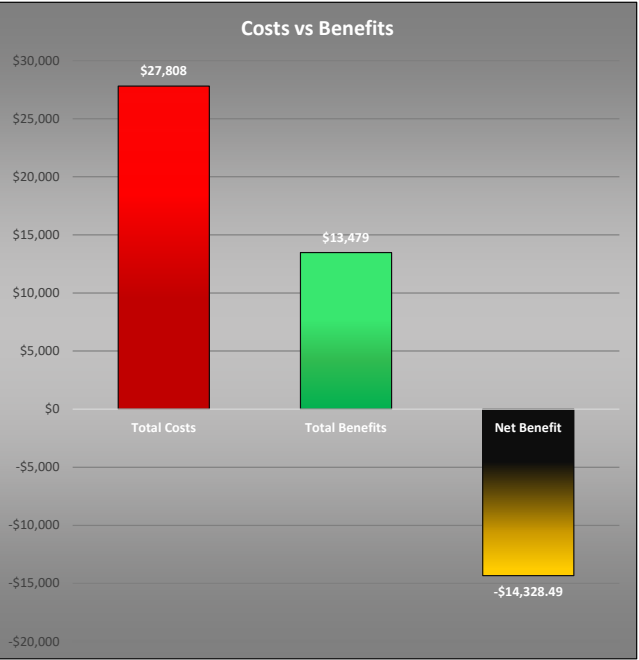
Assumptions and Recommendations

We made several different assumptions in order to calculate the economic costs and benefits. The assumptions include:

- The costs and benefits of the new housing development are solely from the perspective of the city of Cheyenne.
- We assume the houses in the new development are purchased by an equal share of Cheyenne residents and households from outside Laramie County, WY.
- The benefits and costs scale up and down proportional to the size of the housing development.
- The benefits and costs are independent of the location of the housing development.
- The infrastructure, maintenance and water supply costs are covered by charging 1.5 times the water rate of Cheyenne residents.
- The analysis does not consider the option value of leaving the land undeveloped. This may be an important consideration for a city that has limited areas where it can expand to allow for future population growth.

Based on the assumptions above and our sensitivity analysis, our recommendation is for the city of Cheyenne to not provide municipal water to new housing developments in Laramie County. Although we recognize it might be politically complicated, an alternative solution is to annex the land occupied by the new housing development into the city of Cheyenne. In this way, the costs to the city of Cheyenne could be covered by the additional tax revenues, but Laramie County would receive the benefits of a larger population.

Costs	
Category	Value
Expansion of Municipal Water Infrastructure	\$0
Stress on Water System/Supply	\$0
Stress on Other City Services	\$14,335
Decrease in 4% Sales Tax	\$6,945
Decrease in 1% Sales Tax	\$5,631
Decrease in Gas Tax	\$10
Decrease in Severance Tax	\$53
Decrease in Direct Distribution	\$833
Total Cost	\$27,808



Benefits	
Category	Value
Increase in 4% Sales Tax	\$7,265
From Home Construction	\$4,535
From New Residents	\$2,731
Increase in 1% Sales Tax	\$5,993
From Home Construction	\$3,779
From New Residents	\$2,214
Increase in Gas Tax	\$221
Total Benefit	\$13,479

Net Benefit = Total Benefit - Total Costs	-\$14,328.49
Net Benefit Per Development Resident	-\$625.70

Parameters	
Category	Value
Migration Percentage from City Limits	50%
People Per Household	2.29
Acreage for Development	5
Number of Houses on 5 Acres	10
Number of Houses in the Development	10

No Municipal Water

Provide Municipal Water

Ex Ante	Population of Cheyenne	65,132
	Population of Unincorporated Laramie Co.	33,683
	Total Population of Laramie Co.	100,512
	Population of Incorporated Wyoming	398,373
	Albin, Burns, and Pine Bluffs	
Constant	Population	1,697
Ex Post	Population of Cheyenne	65,121
	Population of Unincorporated Laramie Co.	33,706
	Total Population of Laramie Co.	100,523
	Population of Incorporated Wyoming	398,362

Parameters		
Migration Coefficient		100
Acerage for Development		5
People per Household		2.29
New Residents in Development		22.9
Houses in Development		10
Number of Houses on 5 Acres		10
Construction Inputs Value		\$58,334
Per HH Expenditures		\$1,434
Total Scaling Expenditures		\$40,771,676
Households in Cheyenne		28,442

Known Values (Ex Ante)		
Total 4% Sales Tax Revenue		\$37,000,000
Total 1% Sales Tax Revenue		\$30,000,000
Per Capita 4% Sales Tax		\$368.12
Per Capita 1% Sales Tax		\$298.47
Cheyenne's Percent of County Population		64.80%
Total Gas Tax Revenue		\$1,590,000
Per Capita Gas Tax		\$24.41
Total Severance Tax		\$2,200,000
Per Capita Severance Tax		\$33.78
Total Direct Distribution to Cheyenne		\$4,775,250
Inferred Value from Madden/Bebout		\$72.78

Estimated Values (Ex Post)		
Increase in 4% Sales Tax Revenue		\$4,215
Increase in 1% Sales Tax Revenue		\$3,418
Cheyenne's Percent of County Population		64.78%
New Total Gas Tax Revenue		\$1,590,280
Increase in Total Gas Tax Revenue from Distributor Sales		\$209.64
Increase in Total Gas Tax Revenue from Population		\$70